
EDUCATION

Sant'Anna School of Advanced Studies

Sep 2022 – Exp. Nov 2027

*Master's degree – Aerospace engineering - Analytical and Computational Methods for Engineering**Pisa, Italy*

- Honor's student, five-year full scholarship, 5% admission rate
- **Relevant Coursework:** CFD & FEM Numerical Methods, Statistical Data Analysis, Rarefied Gas & Cold Plasma Dynamics, Autonomous Flight (ROS2, PX4), Microcontroller Programming in C, Signal Processing
- **Academic Projects:**
 - **Plasma source characterization** *(in progress)*
 - **Air-breathing electric thruster CubeSat concept** for Very Low Earth Orbit missions *(in progress)*
 - **Godunov-type Riemann solver implementation** in Python to solve hyperbolic PDEs for compressible CFD
 - **3D transient FEA modelling in ANSYS APDL** to simulate moving Hertzian contact loads, coupled with a Python-based spectral diagnostic index to detect structural failures in **rail joints**

University of Pisa

Sep 2022 – Exp. Nov 2027

Master's degree – Aerospace engineering

- **GPA 29.8/30**

- **Relevant Coursework:** Experimental Aerodynamics, Flight Mechanics, Gas Dynamics, Control Systems, Aerospace Structures

Bachelor's degree – Aerospace engineering

Sep 2022 – Jul 2025

- **Grade: 110/110 Summa cum Laude** (Top 1% of class, GPA 29.53/30)
- **Relevant Coursework:** Fluid Dynamics, Continuum Mechanics, Aerospace Systems, C/C++ Programming
- **Academic Projects:** Conducted Star-CCM+ CFD/CHT simulations of an axial compressor stage and modelled multi-body landing gear dynamics in MATLAB.

Lead The Future Mentee

- Among the few Italian students selected to be mentees for **LeadTheFuture**, a leading mentorship non-profit organization for students in STEM, with acceptance rate below 20%.

EXPERIENCE

Caelus Sant'Anna

Autonomous quadcopter team | *CAD and Additive Manufacturing Engineer*

Jan 2023 – Sep 2023

- Developed and implemented the design and additive manufacturing of the structural framework for the main sensor array.
- Secured **1st place worldwide** at the IEEE UAV Challenge 2023 (USA)

Fixed-Wing UAV Team | Co-Founder, Aerodynamics Division Manager & Lead Operations

Sep 2023 – Sep 2025

- **Co-founded** the team to design and build an autonomous fixed-wing UAV from scratch for the **IMechE UAS Challenge** (UK)
- Overlooked Aerodynamic design, systems integration, and autonomous navigation and coordinated design and manufacturing phases to ensure on-time project delivery within tight deadlines
- Designed the wing, fuselage, and A-tail pusher configuration for a new UAV platform.
- Implemented a **parametric simulation framework**, allowing for **fast design iterations** by **automating** geometry changes, mesh generation, and solver configuration. This reduced turnaround time and enabled consistent testing across different design variants.

SkyEye Systems | Remote

BS Thesis Researcher

Apr 2025 – Jun 2025

- Designed and evaluated rotor aerodynamic performance for a large-scale VTOL UAV as part of a bachelor's thesis collaboration with the Laboratory of Experimental Aerodynamics (UniPi).
- Fully automated the simulation pipeline to generate aerodynamic maps and identify the most efficient configuration.

E-Team Squadra Corse | Formula Student Team

CFD Engineer

Oct 2024 – Sep 2025

- Contributed to the setup of **cornering simulations** by implementing a rotating circular domain in Star-CCM+, enabling more realistic modelling of airflow
- Independently handled the validation of CFD results against wind tunnel data by **designing** the test model, **conducting** the trials, and performing **statistical signal analysis on time-series sensor data** in Python.
- **Delivered the engineering design** presentation at **Formula Student Germany 2025**

Head of Aerodynamics

Sep 2025 – Sep 2026

- **Led** a 13-member division to deliver a 13.5% downforce increase, 16% efficiency gain, and 7.3% higher track score.
- Pioneered the team's first **CFD/CHT** battery cooling simulations, lowering steady-state peak temperatures by 15°C. (60°F)
- Optimized simulation architecture to cut standard deviation 10-fold while lowering cell count
- Automated volume meshing and aerodynamic map generation pipelines.
- **Developed a Reduced Order Model** for vehicles dynamics to qualify aerodynamic designs based on predicted track scores
- Developed **experimental validation** via on-track pushrod extensometry and wind tunnel testing
- Led end-to-end composite manufacturing (mould tooling, carbon-fibre layup, curing) to cut package weight by 8%

VOLUNTEERING & ASSOCIATIONS

Alumnus – Accademia Giovani per la Scienza

Aug 2022 – Present

- Deliver talks and presentations aimed at inspiring high school students to explore careers in STEM, fostering curiosity and encouraging early engagement in these fields.

CERTIFICATIONS, SKILLS & INTERESTS

- **Languages:** Italian (Native), English (C2 Cambridge Certificate in Advanced English), French (Conversational), German (Basic)
- **CAE Software:** Star-CCM+, OpenFoam, ANSYS (Fluent, MAPDL, Mechanical, ACP, Workbench), BetaCAE (ANSA, META)
- **CAD & Composites:** Siemens NX, Fibersim, SolidWorks, Autodesk Fusion 360
- **Programming & Scripting:** Python, MATLAB, C, C++, APDL, Star-CCM+ Java Macro Scripting, Git
- **Technical Skills:** Statistical Data Analysis, Team and Project Management, Computational Fluid Dynamics, FEM, composite materials, structures and layups, advanced additive manufacturing (FDM, SLM)
- **Interests:** Photography, Piano, Guitar, University Choir, Hiking